

any delay. Low latency makes the network less prone to other network signals. According to Nokia analytics, its latency response time is as less as 3 milliseconds, while 4G LTE's response fluctuates at 8-10 milliseconds. This makes 5G easy to integrate in the upcoming intelligent devices, making transmission and reception of log data and command sets easier.

The IEEE organisation has standardised this technology so that scientists and researchers from all over the world can access it and utilise accordingly (refer box and Fig. 1).

The 1Gbps data rate of 5G networks also seems to support all the augmented and mixed reality based applications 'live' on the Internet. It will stream multimedia at much higher data rate, so users will be able to run 4K ultra-HD videos very easily.

MIT is doing its own tech analysis and researches based on the 5G telecom techniques and improving algorithms for transferring digital frames in patterned ways. According to some analysts, by 2020, 5G services will start in India under a public-private partnership project (5G PPP).

5G makes its way in medical sector and very stringent areas also, since the full-duplex transceiver provides a very accurate and fast response. Some militaries already use full-duplex technology, which relies on bulky equipment. To achieve full

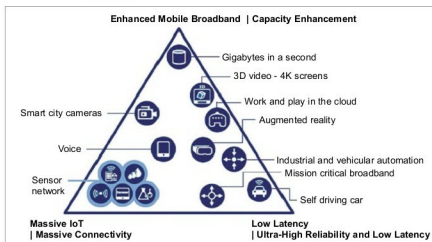


Fig. 1: Miscellaneous IEEE standards (Source: <https://5g.ieee.org/>)

duplex in personal devices, researchers must design a circuit that can route incoming and outgoing signals such that these don't collide when an antenna transmits and receives data at the same time.

This is especially difficult because of the radio waves' tendency to travel both forward and backward on the same frequency—a principle known as reciprocity. But recently, experts have assembled silicon transistors that act like high-speed switches to halt the backward roll of these waves, enabling them to transmit and receive signals on the same frequency at once.

One drawback of full-duplex technology is that it also creates more signal interference through

a pesky echo. When a transmitter emits a signal, that signal is much closer to the device's antenna and therefore more powerful than any signal it receives. Expecting an antenna to both speak and listen at the same time is possible only with special echo-canceling technology.

With these and other 5G technologies, engineers hope to build the wireless network that future smartphone users, virtual reality gamers and autonomous cars will rely on every day. Already, researchers and companies have set high expectations for 5G by promising ultra-low latency and record-breaking data speeds for consumers. If they can solve the remaining challenges, and figure out how to make all the systems work together, ultra-fast 5G service could reach consumers in the next five years.

The Industrial IoT (IIoT) requires highly confidential, reliable and totally secured connectivity. Experts feel that 5G provides the perfect platform for the IIoT.

To conclude

There is a lot more happening in the 5G networks space. While 6G and 7G future generations are already under research in some parts of Europe and Japan, 5G is creating a lot of hype, and developers are striving to make the most of it. **EFY**

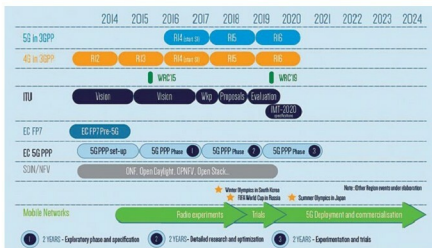


Fig. 2: 5G roadmap (Source: <https://www.quora.com/>)